

Lab Assignment 10

CS 331: Machine Learning

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To be done individually.

The purpose of this lab is to understand the bias variance tradeoff through simulations. We will consider the following hypothesis classes.

- $\mathcal{H}_1$: Constant hypothesis class. The learnt function will be $g^{\mathcal{D}}(x) = \sum_{(x_i, y_i) \in \mathcal{D}} y_i / |\mathcal{D}|$.
- $\mathcal{H}_2$: Linear hypothesis class. The learnt function will be $g^{\mathcal{D}}(x) = wx + b$ where w and b are to be estimated from the dataset $\mathcal{D}$ using linear regression. You may use sklearn's linear regression module for learning w and b from a dataset $\mathcal{D}$.

Perform the following tasks.

1. Write a function to generate a dataset of n points for regression where $y = \sin(\pi x) + \epsilon$, $x \in [-1, 1]$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Several calls will need to be made to this function.
2. For each hypothesis class, compute the average predictor function $\bar{g}(x)$.
3. Generate 100 datasets $\mathcal{D}$ containing $n = 2$ noiseless points ($\epsilon = 0$). Plot the learnt functions and average predictor function $\bar{g}(x)$ for both $\mathcal{H}_1$ and $\mathcal{H}_2$.
4. Generate 100 datasets $\mathcal{D}$ containing $n = 2$ noisy points ($\epsilon \neq 0$). Plot the learnt functions and average predictor function $\bar{g}(x)$ for both $\mathcal{H}_1$ and $\mathcal{H}_2$.
5. Compute the bias, variance and overall error of $\mathcal{H}_1$ and $\mathcal{H}_2$ using simulations. Recall that

$$\begin{aligned}\text{Bias}(\mathcal{H}) &= \mathbb{E}_x [(\bar{g}(x) - f(x))^2] \\ \text{Var}(\mathcal{H}) &= \mathbb{E}_x \mathbb{E}_{\mathcal{D}} [(g^{\mathcal{D}}(x) - \bar{g}(x))^2] \\ \text{Overall Error}(\mathcal{H}) &= \mathbb{E}_x \mathbb{E}_{\mathcal{D}} [(g^{\mathcal{D}}(x) - y)^2]\end{aligned}$$

To do this, you will need to generate multiple training datasets as well as multiple test datasets.

6. Verify the following for datasets of various sizes n .

$$\text{Overall Error}(\mathcal{H}) = \text{Bias}(\mathcal{H}) + \text{Var}(\mathcal{H}) + \text{Noise}$$

where $\text{Noise} = \mathbb{E}_x (y - f(x))^2$.

7. For each of $\mathcal{H}_1$ and $\mathcal{H}_2$ generate a figure containing the learning curves E_{in} vs n , E_{out} vs n and average bias vs n . You may use noiseless datasets.
8. Briefly report your observations for the following cases in a text file or pdf.
 - (a) For $n = 2$ noiseless, which hypothesis class is better in terms of bias, variance and overall errors?
 - (b) When the number of points changes from $n = 2$ to $n = 10$ how do the bias and variance of each hypothesis class change?
 - (c) For each hypothesis class, compare $n = 2$ noiseless vs $n = 2$ noisy with respect to bias and variance.